Synthesis of Hexyl Caproate via Fischer Esterification

Aidan Leonard, Dr. Kawagoe, Dr. MacPherson

03/27/26

1 Introduction

The objective of this experiment is to synthesize hexyl caproate via Fischer Esterification by reacting 1-hexanol with caproic acid in the presence of catalytic concentrated sulfuric acid. The product will be characterized and analyzed by odor analysis, ATR-IR, 100 MHz ^{1}H NMR, and GC-MS.

2 Reaction Scheme

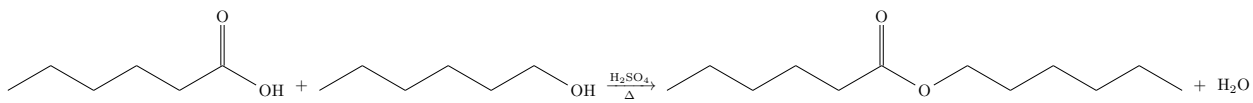


Figure 1: Reaction of Caproic Acid and 1-Hexanol

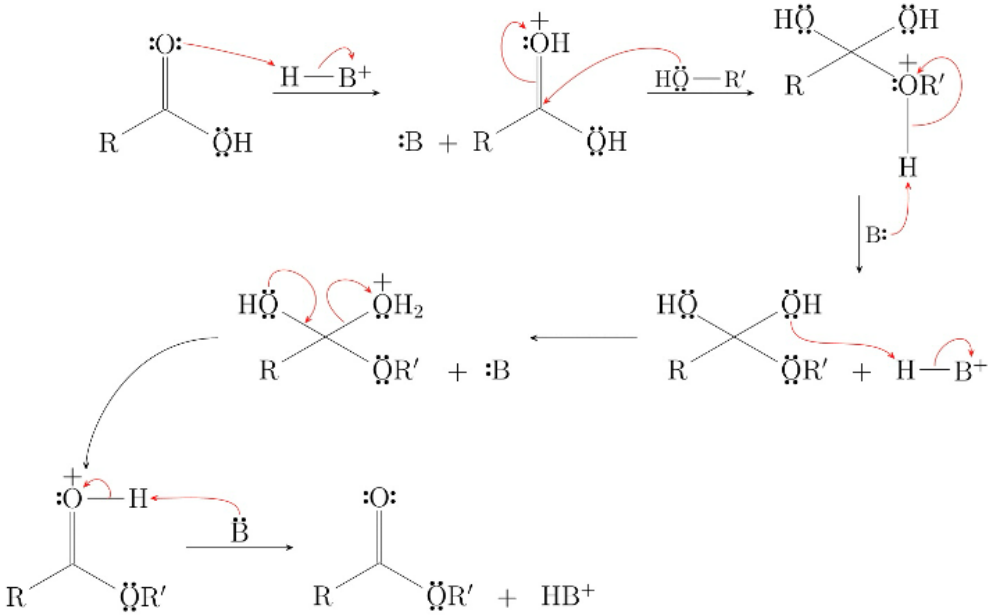


Figure 2: Fischer Esterification Mechanism

3 Table of Reagents and Products

Compound	MW (g/mol)	Density (g/mL)	B.P (°C)	Equiv	Amount	mmol
Caproic Acid	116.16	0.93	202 - 206	1.0	3.49 g / 3.75 mL	30.0
1-Hexanol	102.17	0.82	157	1.3	3.99 g / 4.87 mL	39.0
Sulfuric Acid	98.08	1.84	–	cat.	3 drops	–
Toluene	92.14	0.867	110.6	–	43.35 g / 50 mL	–
Hexyl Caproate	200.32	0.86	245	–	6.02 g / 7 mL	30.0

4 Experimental Procedure

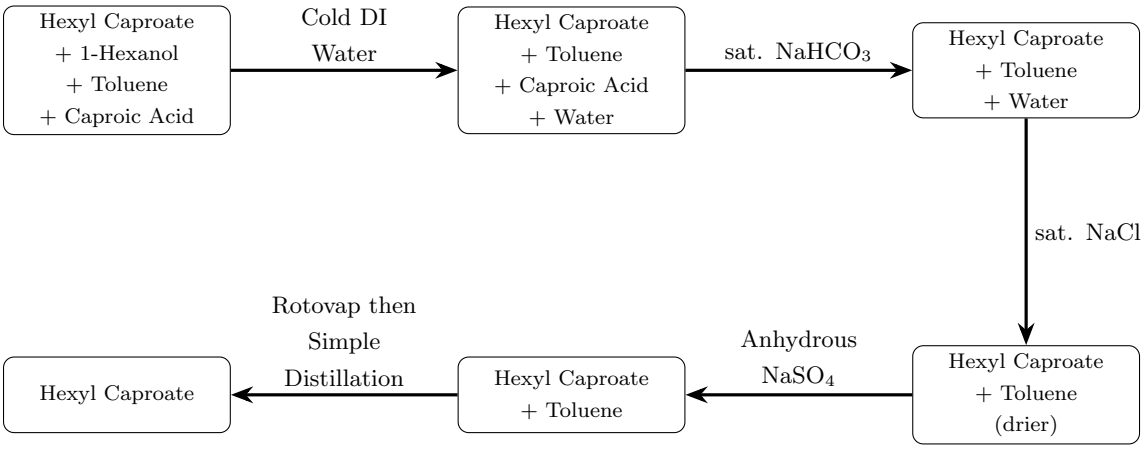
4.1 Apparatus

- Dean Stark Trap
- Simple Distillation
- Rotary Evaporator

4.2 Procedure

- In a 150 mL round-bottom flask, add 3.49 g (3.75 mL) of caproic acid followed by 3.99 g (4.87 mL) of 1-hexanol and a boiling bead.
- While stirring, add 3 drops of concentrated sulfuric acid.
- Add 50 mL of toluene to the flask and set up the dean stark apparatus.
- Gently reflux until around 0.5 mL of water are collected, then allow the mixture to cool to room temperature.
- Chill the product in an ice bath then transfer the mixture to a separatory funnel.
- Quench the organic layer with 15 mL of iced cold DI water and collect the organic layer. Do this twice.
- Neutralize and extract the organic layer with small portions of a saturated sodium bicarbonate solution until the production of CO_2 gas ceases.
- Wash the remaining organic layer with a brine solution then dry the organic over anhydrous sodium sulfate followed by gravity filtration.
- Remove the toluene in the rotatory evaporator at 300 mbar and 50°C water bath to obtain the crude hexyl caproate.
- Purify the product with simple distillation collecting the fraction from 200°C - 250°C.
- Analyze the purified product by odor assessment, ATR-IR, ^{1}H NMR (100 MHz), and GC-MS to confirm ester formation and purity.

5 Purification Scheme



6 Results and Data

g of Caproic acid	g of 1-Hexanol	Drops of H_2SO_4	Reflux Time
Mass of Crude Product		Mass of Purified Product	
Theoretical Yield			

Crude Percent Yield	Pure Percent Yield
Crude Product Observations	Purified Product Observations